Yuheng Wang

Undergraduate in Engineering Physics
College of Engineering
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Education

- **University of Michigan** Sep. 2024 - Current (Expected to graduate in May 2026)
  Undergraduate in Engineering Physics, minor in Computer Science Ann Arbor, USA
  - GPA: 3.96/4.00
- **Shanghai Jiao Tong University** Sep. 2022 - Current (Expected to graduate in July 2026)
  Undergraduate in Electrical and Computer Engineering Shanghai, China
  - GPA: 3.77/4.00

Research Interest

I am broadly interested in the intersection of physics and computation. My primary focus lies in quantum physics and non-Hermitian photonics, with an emphasis on understanding fundamental physical principles through simulation and modeling. Leveraging my dual background in Engineering Physics and Electrical & Computer Engineering, I am also interested in developing and applying artificial intelligence and machine learning methods to accelerate scientific discovery, such as enhancing physical simulations, analyzing complex experimental data, and bridging theoretical models with experimental observables.

Research Experience

- **University of Michigan, Department of Radiation Oncology** May. 2025 - May. 2026
  Undergraduate Research Assistants Ann Arbor, USA
  - Advisor: Lise Wei
  - Equating the Biologically Effective Dose of SBRT and 90Y SIRT: A Unified Voxel-Level Liver Function Dose–Response
    * Built a voxel-level pipeline aligning DGAE-MRI $\mathrm{k}_1(x)$ maps to CT-based dosimetry; defined parenchyma masks and dose binning to quantify $\%\Delta k_1$ post-therapy.
    * Harmonized modalities onto a common biological scale: LQ-based EQD2 for SBRT and protraction-aware BED (Lea–Catcheside kernel) for SIRT.
    * Modeled subvoxel SIRT nonuniformity via EUD, then fit mixed-effects 4-parameter logistic dose–response with covariates; calibrated repair rate μ and micro-dose CV c by aligning SBRT/SIRT curves with bootstrap uncertainty.
  - ReMasker-Cox: Survival-Guided Masked Autoencoder for Missing Data Imputation
    * Extended a masked autoencoder (ReMasker) to jointly optimize imputation and survival consistency by adding a CoxPH loss (PyCox) to the reconstruction objective.
    * Implemented a training workflow with random masking, joint loss weighting (imputation + Cox), and evaluation focused on preserving risk structure for clinical tabular data.
- **University of Michigan, Deng Research Lab** Sep. 2024 - Aug 2025
  Undergraduate Research Assistants Ann Arbor, USA
  - Advisor: Chulwon Lee, Faculty Instructor: Hui Deng
  - Loss–Induced Synthetic Magnetic Field
    * Designed & simulated rotationally asymmetric Si ring–resonator arrays to engineer loss–induced synthetic magnetic fields; observed two-sided non-Hermitian skin-effect localization in 3D FDTD.
    * Built a non-Hermitian tight-binding model; eigen-analysis identified zero net loop phase as the barrier to skin-effect; proposed phase/loss asymmetry and distance-offset remedies and validated designs in simulation.

    - ToolsANSYS Lumerical FDTD, MATLAB; GrateLake HPC.
  - Non-Hermitian Exciton-Polariton Transport beyond Lorentz Reciprocity Theorem
    - Fall 2024 - Winter 2025 Undergraduate Research Opportunity Program
    - Using Ansys Lumerical to simulate a nanobeam cavity integrated with a grating coupler
    - Exfoliating MoSe2 to make monolayer sample
    - Manuscripts submitted: Lee, C., Lee, J.H., Zhang, K., Wang, Y., **Wang, Y.**, Lydick, N., Miao, J., Xin, X., Liu, Y., Sun, K., & Deng, H. "Harnessing nanophotonic edgemodes by engineering PT symmetry"

- **Shanghai Jiao Tong University, Laboratory of Ultrafast Integrated Systems (LUIS)** Aug. 2023 - May 2024
  Undergraduate Research Assistants Shanghai, China
  - Advisor: Xuyang Lu
  - Laboratory website: https://sites.gc.sjtu.edu.cn/xuyang-lu/
  - ENGR4903J Spring 2024: Chip-scale switch-mode wireless power transfer system based on parity-time method
    - Use the Kuramoto model and Adler equation to design an array of oscillators that can change the mutual phase difference by adjusting the coupling coefficient
    - Understand the basic usage of Cadence and the methods associated with analog circuits
  - ENGR4903J Fall 2023: Machine-learning method for mmWave based voice and vital signal recognition
    - Implement the VMD-HHT algorithm to filter noise better and reconstruct the received signal
    - Learn the underlying principles of microcontroller systems, how signals are sent and received between microcontrollers

Internship

- **K-free Technology Co., LTD.** August. 2023 - October. 2023
  Internship, Department of Software Shenzhen, China
  - Responsible for implementing new features in single-chip microcomputers and other system coding tasks in C.
  - Get familiar with the company's code specifications and professional development environment

Teaching and Mentoring Experience

- **JI Center for Learning and Teaching, Shanghai Jiao Tong University** Sep. 2023 - Aug. 2024
  ENGR151(Accelerated Introduction to Programming), ECE311(Analog Circuits) Teaching Assistant Shanghai, China
  - Hold RC to students as ENGR151 Teaching Assistant
  - Hold Lab to students as ECE311 Teaching Assistant
  - Contribute to the scripts for the teaching team to automate workflows

Projects

- **Coupled Flow Matching (CPFM): Controllable Dimensionality Reduction & Reconstruction** May. 2025 - Oct. 2025
  https://arxiv.org/abs/2510.23015 Ann Arbor, USA
  - CPFM, a framework that learns coupled continuous flows for data x and low-dim embeddings y to enable both $p(y|x)$ sampling (latent-space flow) and $p(x|y)$ reconstruction (data-space flow).
  - Extended Gromov–Wasserstein OT with a kernelized quadratic objective to encode semantic priors; derived an alternating minimization with entropic Sinkhorn, achieving $O(n^2)$ per-iteration complexity with an ε-scheduling scheme for stability.
  - Implemented Dual Conditional Flow Matching (DCFM) with a shared drift network, role flagging, and mute-masking to jointly learn bidirectional flows; built a voxel-to-latent pipeline for 2D embeddings and high-fidelity reconstructions.
  - Evaluated on MNIST, CIFAR-10, TinyImageNet, AFHQ, and QM9; obtained semantically rich 2D embeddings and higher-fidelity reconstructions than VAE/DiffAE/Info-Diffusion under extreme compression.
- **Curie: Automate Rigorous Scientific Experimentation** Mar. 2025 - Aug. 2025
  https://github.com/Just-Curieous/Curie Ann Arbor, USA
  - Testing and developing new features
- **Analogical analyses of epidemic evolution from the fluid mechanics view** Sep. 2022 - May. 2024
  The 26th 'Shanghai Jiaotong University Student Innovation and Practice Programme' Project Shanghai, China
  - Model epidemic evolution using meta cellular automate written in Python and Matlab
  - Study second-order phase transitions and link to outbreaks of epidemics

Technical Skills

Languages: C, C++, Python, Elm, Bash
Miscellaneous: LaTeX, Scratch, Verilog, Russian

Achievements

2025	**Michigan Engineering Continuing Student Scholarship (Roger King Scholarship)**
2025	**Umich University Honors 2024-2025**
2025	**Umich Dean's List 2024-2025**
2024	**SJTU Undergraduate Excellent Scholarship for the Academic Year 2023-2024**
2024	**Meritorious Winner**, 2024 COMAP's Mathematical Contest in Modeling (MCM)
2023	**SJTU Undergraduate Excellent Scholarship for the Academic Year 2022-2023**
2020	**First Prize**, Chinese Physics Olympic (CPhO) of Guangdong province